

Lukas Jander

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Education

Rochester Institute of Technology, GPA: 3.99/4.00, Dean's List, Honors Program Rochester, NY
BS/ME Mechanical Engineering Expected Graduation May, 2027

Experience

Laboratory for Laser Energetics Engineering Intern January, 2025 – July 2026

- Designed and analyzed a flexure-based 3 degree of freedom mirror mount
 - Used Monte Carlo simulation (ANSYS + Sigfit) to quantify sensitivity of optical performance to mounting point tolerances
 - Verified diffraction limited performance at 6THz via raytracing on deformed optical surfaces
 - Ran ANSYS parametric studies to select materials and dimensions to minimize surface RMS
- Redesigned a vacuum compatible lens assembly to accommodate two optic sizes (6" and 150mm)
 - Eliminated metal on optic contact using moveable Delrin pins
 - Performed tolerance stack up to verify positioning accuracy
- Diagnosed a temperature gradient issue in an existing LN₂ driven argon gas cooling system
 - Measured temperature profiles to estimate convective heat transfer coefficients
 - ANSYS thermal simulations compared favorably to experimental data within 3K
- Iterated a rigging fixture design for a 60lb X-ray camera following an equipment damaging incident
 - Diagnosed cause of failure and evaluated multiple potential solutions
 - Validated redesign with ANSYS mechanical and bolted joint hand calculations
 - New design is used on multiple other cameras after successfully being deployed on X-ray camera
- Built a passive lift mechanism to reduce the perceived lift weight of a 35lb calorimeter to ~5lb
 - Ensured the design's safety and ease of use based on physical prototyping

Insulet Corporation Quality Engineering Intern January, 2026 – April 2026

- Wrote test procedures for cyclic load testing of a medical device
 - Programmed Instron tensile test methods to automate testing
- Created torsional test setups to evaluate repeatability of manufacturing procedures
- Characterized repeatability of surface roughness measurements to investigate ink adhesion issues

SEPAC Inc. Engineering Intern May, 2024 – August, 2024

- Designed a test setup to automate brake performance evaluation
 - Integrated a torque supplying motor with transducer feedback and PLC control
- Wrote programmable logic controller (PLC) code to automate test procedures
- Performed hardware and software repairs on existing test setups to minimize downtime

Rochester Institute of Technology Researcher in a Microelectronics Lab May, 2023 – August, 2023

- Characterized sensitivity of five different designs of a MEMS viscosity sensor to monitor oil health
- Created a high pass filtering and amplification circuit to remove DC offset and low frequency noise
- Isolated and subtracted out temperature dependent sensor drift
- Findings presented in *Waterproofing a Thermally-Actuated Vibrational MEMS Viscosity Sensor*

Projects and Clubs

RIT Competitive Rocketry Team: Airbrakes Subteam Member

- Performed FEA to evaluate mechanical load cases experienced during flight
- Designed an active apogee control mechanism to meet target apogee
 - Wrote MATLAB scripts to predict load on components and optimize design
- 2025-26 rocket successfully flew within 1.8% of target apogee of 10,000ft using this mechanism
- Performing computational fluid dynamics on a computing cluster to obtain drag estimates for the 2026-27 rocket

Combination Lock Solver

- Designed and built a device that opens combination locks and display the solution
- Arduino controlled stepper motor, solenoid, and a beam break sensor drove the mechanism
- Testing of servo motor and design of PCB in progress

Treasurer of Rock Climbing Club

- Organized events, fundraised, and communicated with university staff about budget

Skills

Software: ANSYS, MATLAB, SigFit, LabVIEW, NX, SOLIDWORKS, Creo, Autodesk Inventor/Fusion

Miscellaneous: GD&T proficient, Fluent in German and Czech